Sample Campus Network

Coursework Brief

Albion University is a large university which has two campuses situated 20 miles apart. The university's students and staff are distributed in 4 faculties; these include the faculties of Health and Sciences; Business; Engineering/Computing and Art/Design. Each member of staff has a PC and students have access to PCs in the labs.

Requirements:

a. Create a network topology with the main components to support the following:
  - Main campus:
    - **Building A**: Administrative staff in the departments of management, HR and finance. The admin staff PCs are distributed in the building offices and it is expected that they will share some networking equipment (**Hint: use of VLANs is expected here**). The Faculty of Business is also situated in this building
    - **Building B**: Faculty of Engineering and Computing and Faculty of Art and Design
    - **Building C**: Students' labs and IT department. The IT department hosts the University Web server and other servers
    - There is also an email server hosted externally on the cloud.
  - Smaller campus:
    - Faculty of Health and Sciences (staff and students' labs are situated on separate floors)

b. You will be expected to configure the core devices and few end devices to provide end-to-end connectivity and access to the internal servers and the external server.
  - Each department/faculty is expected to be on its own separate IP network
  - The switches should be configured with appropriate VLANs and security settings
  - RIPv2 will be used to provide routing for the routers in the internal network and static routing for the external server.
  - The devices in building A will be expected to acquire dynamic IP addresses from a router-based DHCP server

Tasks:

Task 1: Your task is to plan, design, and prototype the network topology for Albion University's network using Cisco Packet Tracer. **Formative feedback will be given on this task in week 6.**

Task 2: Configure in Packet Tracer the network with appropriate settings to achieve the connectivity and functionalities specified in the requirements.

Task 3: Produce a report (**max 1500 words**) including evaluation your proposed network design and critical appraisal on your work. Your evaluation should include performance, scalability, reliability and security of your proposed network.

Main-Campus-Router

```
Main-Router(config-if)#int se0/3/1
Main-Router(config-if)#clock rate 64000
Main-Router(config-if)#int se0/3/0
Main-Router(config-if)#clock rate 64000
Main-Router(config)#int se0/3/1
Main-Router(config-if)#ip address 10.10.10.1 255.255.255.252
Main-Router(config-if)#int se0/3/0
```

```
Main-Router(config-if)#ip address 10.10.10.5 255.255.255.252
Main-Router(config)#int gig0/0.10
Main-Router(config-subif)#encapsulation dot1Q 10
Main-Router(config-subif)#ip address 192.168.1.1 255.255.255.0
Main-Router(config-subif)#exit
Main-Router(config)#int gig0/0.20
Main-Router(config-subif)#encapsulation dot1Q 20
Main-Router(config-subif)#ip address 192.168.2.1 255.255.255.0
Main-Router(config-subif)#exit
Main-Router(config)#int gig0/0.30
Main-Router(config-subif)#encapsulation dot1Q 30
Main-Router(config-subif)#ip address 192.168.3.1 255.255.255.0
Main-Router(config-subif)#exit
Main-Router(config)#int gig0/0.40
Main-Router(config-subif)#encapsulation dot1Q 40
Main-Router(config-subif)#ip address 192.168.4.1 255.255.255.0
Main-Router(config-subif)#exit
Main-Router(config)#int gig0/0.50
Main-Router(config-subif)#encapsulation dot1Q 50
Main-Router(config-subif)#ip address 192.168.5.1 255.255.255.0
Main-Router(config-subif)#exit
Main-Router(config)#int gig0/0.60
Main-Router(config-subif)#encapsulation dot1Q 60
Main-Router(config-subif)#ip address 192.168.6.1 255.255.255.0
Main-Router(config-subif)#exit
Main-Router(config)#int gig0/0.70
Main-Router(config-subif)#encapsulation dot1Q 70
Main-Router(config-subif)#ip address 192.168.7.1 255.255.255.0
Main-Router(config-subif)#exit
Main-Router(config)#int gig0/0.80
Main-Router(config-subif)#encapsulation dot1Q 80
Main-Router(config-subif)#ip address 192.168.8.1 255.255.255.0
Main-Router(config-subif)#exit
Main-Router(config)#service dhcp
Main-Router(config)#ip dhcp pool admin-pool
Main-Router(dhcp-config)#network 192.168.1.0 255.255.255.0
Main-Router(dhcp-config)#default-router 192.168.1.1
Main-Router(dhcp-config)#dns-server 192.168.1.1
Main-Router(dhcp-config)#ex
Main-Router(config)#ip dhcp pool hr-pool
```

```
Main-Router(dhcp-config)#network 192.168.2.0 255.255.255.0
Main-Router(dhcp-config)#default-router 192.168.2.1
Main-Router(dhcp-config)#dns-server 192.168.2.1
Main-Router(dhcp-config)#ex
Main-Router(config)#ip dhcp pool finance-pool
Main-Router(dhcp-config)#network 192.168.3.0 255.255.255.0
Main-Router(dhcp-config)#default-router 192.168.3.1
Main-Router(dhcp-config)#dns-server 192.168.3.1
Main-Router(dhcp-config)#ex
Main-Router(config)#ip dhcp pool business-pool
Main-Router(dhcp-config)#network 192.168.4.0 255.255.255.0
Main-Router(dhcp-config)#default-router 192.168.4.1
Main-Router(dhcp-config)#dns-server 192.168.4.1
Main-Router(dhcp-config)#ex
Main-Router(config)#ip dhcp pool e&c-pool
Main-Router(dhcp-config)#network 192.168.5.0 255.255.255.0
Main-Router(dhcp-config)#default-router 192.168.5.1
Main-Router(dhcp-config)#dns-server 192.168.5.1
Main-Router(dhcp-config)#ex
Main-Router(config)#ip dhcp pool a&d-pool
Main-Router(dhcp-config)#network 192.168.6.0 255.255.255.0
Main-Router(dhcp-config)#default-router 192.168.6.1
Main-Router(dhcp-config)#dns-server 192.168.6.1
Main-Router(dhcp-config)#ex
Main-Router(config)#ip dhcp pool studlab-pool
Main-Router(dhcp-config)#network 192.168.7.0 255.255.255.0
Main-Router(dhcp-config)#default-router 192.168.7.1
Main-Router(dhcp-config)#dns-server 192.168.7.1
Main-Router(dhcp-config)#ex
Main-Router(config)#ip dhcp pool itdept-pool
Main-Router(dhcp-config)#network 192.168.8.0 255.255.255.0
Main-Router(dhcp-config)#default-router 192.168.8.1
Main-Router(dhcp-config)#dns-server 192.168.8.1
Main-Router(dhcp-config)#ex
Main-Router(config)#router rip
Main-Router(config-router)#version 2
Main-Router(config-router)#network 192.168.1.0
Main-Router(config-router)#network 192.168.2.0
Main-Router(config-router)#network 192.168.3.0
Main-Router(config-router)#network 192.168.4.0
```

Main-Router(config-router)#network 192.168.5.0
Main-Router(config-router)#network 192.168.6.0
Main-Router(config-router)#network 192.168.7.0
Main-Router(config-router)#network 192.168.8.0
Main-Router(config-router)#network 10.10.10.0
Main-Router(config-router)#network 10.10.10.4
Main-Router(config-router)#ex
Main-Router(config)#do wr

Branch-Router

Branch-Router(config)#int se0/3/1
Branch-Router(config-if)#ip address 10.10.10.2 255.255.255.252
Branch-Router(config)#int gig0/0.90
Branch-Router(config-subif)#encapsulation dot1Q 90
Branch-Router(config-subif)#ip address 192.168.9.1 255.255.255.0
Branch-Router(config-subif)#exit
Branch-Router(config)#int gig0/0.100
Branch-Router(config-subif)#encapsulation dot1Q 100
Branch-Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Branch-Router(config-subif)#exit
Branch-Router(config)#service dhcp
Branch-Router(config)#ip dhcp pool Staff-pool
Branch-Router(dhcp-config)#network 192.168.9.0 255.255.255.0
Branch-Router(dhcp-config)#default-router 192.168.9.1
Branch-Router(dhcp-config)#dns-server 192.168.9.1
Branch-Router(config)#ip dhcp pool Studlb-pool
Branch-Router(dhcp-config)#network 192.168.10.0 255.255.255.0
Branch-Router(dhcp-config)#default-router 192.168.10.1
Branch-Router(dhcp-config)#dns-server 192.168.10.1
Branch-Router(dhcp-config)#ex
Branch-Router(config)#router rip
Branch-Router(config-router)#version 2
Branch-Router(config-router)#network 192.168.9.0
Branch-Router(config-router)#network 192.168.10.0
Branch-Router(config-router)#network 10.10.10.0
Branch-Router(config-router)#ex
Branch-Router(config)#do wr

Cloud-Router

Cloud(config)#int se0/3/0
Cloud(config-if)#ip addr 10.10.10.6 255.255.255.252
Cloud(config-if)#no sh
Cloud(config-if)#ex
Cloud(config)#int gig0/0
Cloud(config-if)#ip addr 20.0.0.1 255.255.255.252
Cloud(config)#router rip
Cloud(config-router)#version 2
Cloud(config-router)#network 10.10.10.4
Cloud(config-router)#network 20.0.0.0
Cloud(config-router)#ex
Cloud(config)#do wr

In Email Server
IP address = 20.0.0.2/30(static)
Subnet mask = 255.255.255.252
Default Gateway = 20.0.0.1

ADMIN-SWITCH

Switch(config)#int range fa0/1-24
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10

HR-SWITCH

Switch(config)#int range fa0/1-24
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20

FINANCE-SWITCH

Switch(config)#int range fa0/1-24
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 30

BUSINESS-SWITCH

Switch(config)#int range fa0/1-24
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 40

E&C-SWITCH

```
Switch(config)#int range fa0/1-24
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 50
```

A&D-SWITCH

```
Switch(config)#int range fa0/1-24
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 60
```

STUD-LAB-SWITCH

```
Switch(config)#int range fa0/1-24
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 70
```

IT-DEPT-SWITCH

```
Switch(config)#int range fa0/1-24
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 80
```

STUD-LB-SWITCH

```
Switch(config)#int range fa0/1-24
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 100
```

STAFF-SWITCH

```
Switch(config)#int range fa0/1-24
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 90
```

Campus-L3-Switch

```
Campus-L3(config)#int gig1/0/2
Campus-L3(config-if)#switchport mode access
Campus-L3(config-if)#switchport access vlan 10
Campus-L3(config)#int gig1/0/1
Campus-L3(config-if)#switchport mode access
Campus-L3(config-if)#switchport access vlan 20
Campus-L3(config)#int gig1/0/3
Campus-L3(config-if)#switchport mode access
Campus-L3(config-if)#switchport access vlan 30
```

```
Campus-L3(config)#int gig1/0/4
Campus-L3(config-if)#switchport mode access
Campus-L3(config-if)#switchport access vlan 40
Campus-L3(config)#int gig1/0/5
Campus-L3(config-if)#switchport mode access
Campus-L3(config-if)#switchport access vlan 50
Campus-L3(config)#int gig1/0/6
Campus-L3(config-if)#switchport mode access
Campus-L3(config-if)#switchport access vlan 60
Campus-L3(config)#int gig1/0/7
Campus-L3(config-if)#switchport mode access
Campus-L3(config-if)#switchport access vlan 70
Campus-L3(config)#int gig1/0/8
Campus-L3(config-if)#switchport mode access
Campus-L3(config-if)#switchport access vlan 80
Campus-L3(config)#int gig1/0/9
Campus-L3(config-if)#switchport mode trunk
```

Branch-L3-Switch

```
Branch-L3-Switch(config)#int gig1/0/1
Branch-L3-Switch(config-if)#switchport mode access
Branch-L3-Switch(config-if)#switchport access vlan 90
Branch-L3-Switch(config)#int gig1/0/2
Branch-L3-Switch(config-if)#switchport mode access
Branch-L3-Switch(config-if)#switchport access vlan 100
Branch-L3-Switch(config)#int gig1/0/3
Branch-L3-Switch(config-if)#switchport mode trunk
```